MLR-DBPFN: A Multi-Scale Low Rank Deep Back Projection Fusion Network for Anti-Noise Hyperspectral and Multispectral Image Fusion

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***Abstract*— Fusing low spatial resolution (LR) hyperspectral (HS) data and high spatial resolution (HR) multispectral (MS) data aims to obtain HR HS data. However, due to bad weather and the aging of sensor equipment, HS images usually contain a lot of noise, e.g., Gaussian noise, strip noise, and mixed noise, which would make the fused image have low quality. To solve this problem, we propose the multiscale low-rank deep back projection fusion network (MLR-DBPFN). First, HS and MS are superimposed, and multiscale spectral features of the stacked image are extracted through multiscale low-rank decomposition and convolution operation, which effectively removes noisy spectral features. Second, the upsampling and downsampling network mechanisms are used to extract the multiscale spatial features from each layer of spectral features. Finally, the multiscale spectral features and multiscale spatial features are combined for network training, and the weight of the noisy spectrum features is reduced through the network feedback mechanism, which suppresses the noisy spectrum and improves the noisy HS fusion performance. Experimental results on datasets of different noise demonstrate that MLR-DBPFN has superior spatial and spectral fidelity, comparative fusion quality, and robust antinoise performance compared with state-of-the-art methods.**

***Index Terms*— Antinoise, deep learning, hyperspectral (HS), image fusion, multiscale low rank (MLR), multispectral (MS).**

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I. Introduction

HYPERSPECTRAL (HS) satellite data have fine spectral resolution and have been widely used in ground objects identification, resource exploration, and environmental monitoring [1], [2]. However, the narrow wave width has limited absorption of reflected energy from the ground surface, and low spatial resolution (LR) is usually obtained, which seriously affects the application fields of satellite HS data [3]. In contrast, MS data obtain higher spatial resolution but coarse spectral information [4]. Accordingly, image fusion that combines the spectral information of LR HS with the spatial information of high spatial resolution (HR) MS is always adopted to obtain HR HS data, which has achieved great popularity in the remote sensing field [5].

In recent years, numerous HS and MS fusion methods have been proposed. These methods can be roughly divided into component substitution (CS)-based methods, multiresolution analysis (MRA)-based methods, iterative optimization-based methods, and deep learning-based methods [6]. The CS-based methods and MRA-based methods are collectively referred to as the hypersharpening methods [7]; these methods use the pansharpening methods for the fusion of HS and MS under the band grouping framework proposed by Chen *et al.* [8] and the band synthesis framework proposed by Selva *et al.* [9]. CS-based methods assume that the spatial resolution of HS data is dominated by the intensity component, and multisource data fusion can be realized by using MS data to replace the intensity component of HS data [10]. Based on the above theory, Carper *et al.* [11] proposed intensity–hue–saturation (IHS) for image fusion by the transformation of different domains. Pal *et al.* [12] used the principal component analysis to achieve multisource image fusion. Aiazzi *et al.* [13] proposed adaptive Gram–Schmidt (GSA) by using multivariate regression and achieved better quality in multisource data fusion. However, these methods ignore the spectral distortion caused by component replacement while improving the spatial resolution of the image [14]. MRA-based methods assume that

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Fig. 1. Noisy HS data.

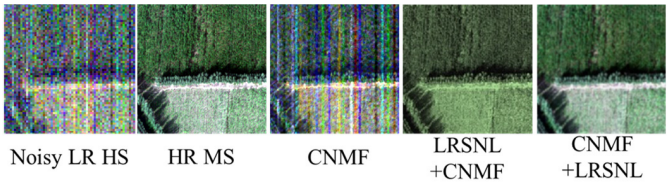


Fig. 2. Influence of noisy data on image fusion.

the HR MS data contain the missing spatial information of LR HS data. Detail information can be extracted by multiresolution decomposition of HR MS data, and the fusion can be realized by injecting the detailed information of HR MS data into LR HS data [15]. Javan *et al.* [16] proposed modulation transfer function-generalized Laplacian pyramid (MTF-GLP) for image fusion. Meng *et al.* [17] used wavelet transform to decompose LR and HR data into multifrequency information and proposed "a Ì trous" wavelet transform (ATWT) to realize image fusion in frequency domain. These methods have better spectral fidelity [18], but the performance of spatial resolution improvement is limited [19]. Iterative optimization-based methods adopt numerous matrix decomposition theories [20]. These methods decompose the HS and MS data and combine the spectral matrix of HS data with the spatial matrix of MS to obtain the fused data [21]. Yokoya *et al.* [22] proposed the coupled nonnegative matrix factorization (CNMF) to fuse HS and MS and achieve comparative fusion performance. Based on the sparse representation theory, Nezhad *et al.* [23] proposed the nonlocal sparse representation method based on pixel group for image fusion. Li *et al.* [24] regarded the remote sensing data as a three-dimensional tensor and proposed the nonlocal sparse tensor factorization (NLSTF). The iterative optimization-based methods always have low computational efficiency, require numerous prior parameters, and have low robustness [10]. Deep learning-based methods have recently been used in the field of image processing [25]. Compared with hypersharpening methods and iterative optimization methods, deep learning-based methods have the advantage of solving nonlinear problems [26]. These methods use prior data to train the model and map the model to experimental data to obtain the fusion result [27]. In order to avoid spatial loss caused by 2-D convolution, Palsson *et al.* [28] proposed 3-D convolutional neural networks (CNNs) for HS and MS data fusion. Yang *et al.* [29] established a spatial feature extraction network and a spectral feature extraction network based on data characteristics and presented two-branch (TWO) CNN. Zhou *et al.* [30] combined the Laplace pyramid decomposition theory with CNN to achieve image fusion on a multifrequency scale. At present, deep learning-based methods have become a hot research field of image fusion due to their performance advantages [31].

Each band of HS sensor has limited absorption of surface reflected energy during imaging [32]. Affected by weather, aging of sensor equipment, and other factors, HS data usually have obvious noise, e.g., Gaussian random noise, stripe noise, and mixed noise (see Fig. 1), which degrades the original spectral properties of ground objects and causes serious errors in Earth observation [3], [33], [34]. Unfortunately, CS-based methods, MRA-based methods, iterative optimization-based methods, and deep learning-based methods mostly focus on the fusion of noise-free HS data and MS data [3]. These methods cause serious spectral and spatial distortions when fusing noisy HS data (see Fig. 2, CNMF) [35]. Some researchers argue that preprocessing of noise reduction can be performed in data fusion. Some HS denoising methods have emerged to solve this problem. For example, Chen and Qian [36] used wavelet decomposition to suppress the HS data noise in the frequency domain. Chen *et al.* [17] used a 3-D nonlinear means filter to reconstruct HS data. Zhu *et al.* [38] proposed a low-rank spectral nonlocal (LRSNL) method for HS denoising. However, this preprocessing procedure may oversmooth spatial details of the fused data, and the error caused by noise reduction would be accumulated in the data fusion to cause more serious spectral distortion (see Fig. 2, LRSNL + CNMF and CNMF + LRSNL). Separating and suppressing noise during the fusion process are crucial to improve the fidelity of the space spectrum.

In this article, we propose a multiscale low-rank deep back projection fusion network (MLR-DBPFN) to fuse noisy HS data with MS data. First, we introduce the MLR decomposition model to extract the multiscale spectral features of the data, which can effectively separate the noise from the image. To ensure that the decomposition scale of LR HS and HR MS data is consistent, LR HS data are interpolated and superimposed with HR MS. Second, a downsampling network and an upsampling network are constructed by using convolution and deconvolution operation, and multiscale spatial features of each spectral feature are extracted. Finally, we combine the multiscale spectral feature extraction model with the multiscale spatial feature extraction network to build MLR-DBPFN. It is worth noting that the feedback mechanism of the network could adaptively reduce the weight of noisy spectral features in image reconstruction and suppress noise. Besides, in order to minimize the spectral and spatial distortions of the fused data, we do not remove the noisy spectral features that contain limited spatial–spectral information, which not only ensures the integrity of the spectral information but also avoids the local error caused by the global spectral features. The main contributions of this article include the following.

1) The proposed MLR-DBPN can effectively decompose and suppress noise when fusing HS and multispectral (MS) images. It extracts multiscale spatial and spectral features, adaptively suppresses noisy spectral features

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in the procedure of network training, and obtains high-quality fusion results of noisy HS data.

2) The MLR-DBPFN makes good use of multiscale spatial–spectral features to promote the data fusion quality. The MLR is used to effectively extract the multiscale spectral features of noisy HS, and the up-down-sampling operation is used to extract the multiscale spatial features of each spectral feature.
3) MLR-DBPFN achieves comparative fusion performance on datasets with noise-free, Gaussian random noise, stripe noise, and mixed noise, and outperforms nine state-of-the-art methods in antinoise behaviors and spatial–spectral fidelity.

The structure of this article is given as follows. Section II introduces the methodology, and experimental results are described in Section III. In the end, conclusions are provided in Section IV.

II. Methodology

In the process of HS imaging, the sensor is affected by factors such as weather and equipment aging, and image noise is definitely generated. How to fuse noisy images is an important but challenging problem. Based on the above motivation, we propose MLR-DBPFN for fusing HS images with different types of noise with MS images. For a better understanding, we use $\mathbf{Z} \in \mathbb{R}^{W\times H\times K}$ for HR HS, $\mathbf{X} \in \mathbb{R}^{w\times h\times K}$ for LR HS, and $\mathbf{Y} \in \mathbb{R}^{W\times H\times k}$ for HR MS.

A. Multiscale Spectral Feature Extraction

Noisy HS images can be regarded as composed of multiscale low-rank components [39]. Therefore, we use the MLR model to decompose the HS data at multiple scales [40]. On the one hand, MLR can separate the pure spectral features from noisy spectral features. On the other hand, a complex spectral feature is decomposed into multiple simple spectral features, and the global spectral feature is transformed into the local spectral feature, which not only obtains richer spectral features but also avoids the local error caused by using global spectral features. Fig. 3 illustrates the multiscale spectral feature extraction results.

To decompose the HS and MS data on the same scale, we interpolate the HS data and superimpose the interpolated HS and MS data, as follows:

$$\mathbf{P} = \text{cat}(\mathbf{X}\uparrow, \mathbf{Y}, 3) \tag{1}$$

where $\uparrow$ represents the interpolation operation, cat(:, :, **3**) represents the superposition of the matrices on the third dimension, and $\mathbf{P}$ represents the data after matrix superposition. Then, we formulate $\mathbf{P}$ into the MLR model

$$\begin{aligned} \mathbf{P}_i &= \sum_{l=0}^{L} \mathbf{P}_i^l \\ s.t.\ \mathbf{P}_i^l &= \sum \mathbf{R}_b^{\mathrm{T}}(\mathbf{B}_l), \quad \mathbf{B}_l = \mathbf{U}_b\mathbf{S}_b\mathbf{V}_b^{\mathrm{T}} \end{aligned} \tag{2}$$

in which $\mathbf{P}_i$ represents the ith band of matrix $\mathbf{P}$ and $L = [l_0, l_1, \ldots, l_n]$ is the number of decomposition layers. $\mathbf{R}_b$ is a block reshape operator; $\mathbf{B}_l$ represents the low-rank component

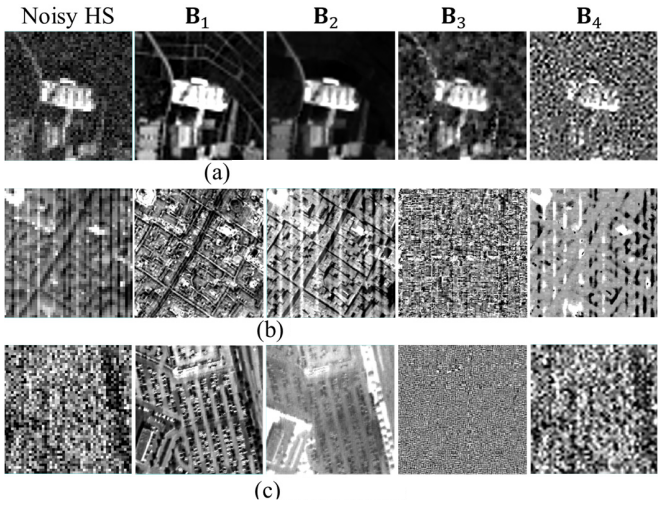


Fig. 3. Results of multiscale spectral feature extraction. (a) Gaussian random noise. (b) Stripe noise. (c) Mixed noise.

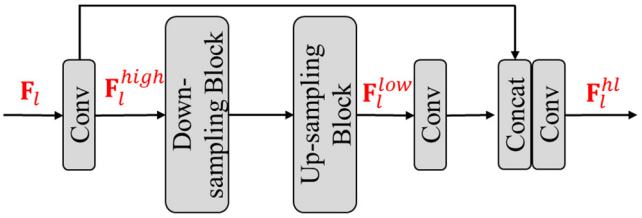


Fig. 4. Multiscale spatial feature extraction process.

of the lth layer; $\mathbf{U}_b$, $\mathbf{S}_b$, and $\mathbf{V}_b$ are resulted from the singular value decomposition (SVD) of $\mathbf{B}_l$; and T is transpose operation. To solve the above formula

$$\begin{aligned} &\arg\min_{\mathbf{p}_i^0, \mathbf{p}_i^1, \ldots, \mathbf{p}_i^L} \sum_{l=0}^{L} \alpha_l \sum_{l=0}^{L} \left\| \mathbf{R}_b\left(\mathbf{p}_i^l\right) \right\|_* \\ &s.t.\ \mathbf{p}_i = \sum_{l=0}^{L} \mathbf{p}_i^l \end{aligned} \tag{3}$$

where $\|\cdot\|_*$ is the nuclear norm operator and α_l is the regularization parameter. The alternating direction method of multipliers (ADMM) is used to solve (3). The multiscale spectral features are acquired

$$\mathbf{F}_l = \text{ReLUConv}(\mathbf{B}_l)) \tag{4}$$

where Conv$(\cdot)$ represents convolution operation, ReLU is the nonlinear activation function of the rectified linear unit, and $\mathbf{F}_l$ is the spectral feature of the lth low-rank component.

B. Multiscale Spatial Feature Extraction

Remote sensing data contain rich spatial information, and the effective use of spatial details can improve the fusion quality of the final images [41]. To use spatial detail information more effectively, we extract the multiscale spatial feature of $\mathbf{F}_l$ [42]. Fig. 4 shows the specific process.

In the process of multiscale spatial feature extraction, $\mathbf{F}_l$ first implements convolution operation to obtain high-frequency features and then combines the downsampling block

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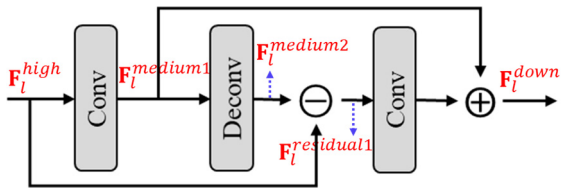


Fig. 5. Process of the downsampling block.

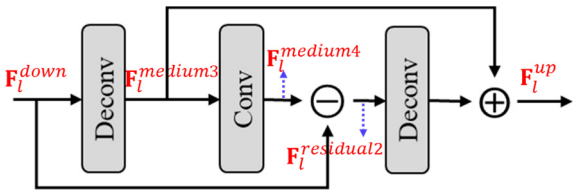


Fig. 6. Process of the upsampling block.

and upsampling block to obtain the low-frequency features of $\mathbf{F}_l$. Finally, the low-frequency features after convolution and high-frequency features are superimposed to obtain the multiscale spatial feature. The specific formula is given as follows:

$$\mathbf{F}_l^{\text{high}} = \text{ReLU}(\text{Conv}(\mathbf{F}_l)) \tag{5}$$

$$\mathbf{F}_l^{\text{low}} = \text{Up} \uparrow \left(\text{Down} \downarrow \left(\mathbf{F}_l^{\text{high}}\right)\right) \tag{6}$$

$$\mathbf{F}_l^{\text{hl}} = \text{cat}\left(\text{ReLU}\left(\text{Conv}\left(\mathbf{F}_l^{\text{low}}\right)\right), \mathbf{F}_l^{\text{high}}, 3\right) \tag{7}$$

where $\mathbf{F}_l^{\text{high}}$ represents the high-frequency feature of $\mathbf{F}_l$, $\mathbf{F}_l^{\text{low}}$ is the low-frequency feature of $\mathbf{F}_l$, and $\mathbf{F}_l^{\text{hl}}$ represents the superposition layer of high- and low-frequency features. Down $\downarrow(\cdot)$ is the downsampling operation, and Up $\uparrow(\cdot)$ is the upsampling operation.

For the down-sampling block [see Fig. 5], we successively perform convolution and deconvolution operations on $\mathbf{F}_l^{\text{high}}$ to, respectively, obtain $\mathbf{F}_l^{\text{medium1}}$ and $\mathbf{F}_l^{\text{medium2}}$. To reduce redundant feature information, $\mathbf{F}_l^{\text{high}}$ is used to subtract $\mathbf{F}_l^{\text{medium2}}$ and obtain the residual features ($\mathbf{F}_l^{\text{residual1}}$). The residual features are finally convolved and added into $\mathbf{F}_l^{\text{medium1}}$ to obtain the downsampling features ($\mathbf{F}_l^{\text{down}}$). The specific algorithm is as follows:

$$\mathbf{F}_l^{\text{medium1}} = \text{ReLU}\left(\text{Conv}\left(\mathbf{F}_l^{\text{high}}\right)\right) \tag{8}$$

$$\mathbf{F}_l^{\text{medium2}} = \text{ReLU}\left(\text{Deconv}\left(\mathbf{F}_l^{\text{medium1}}\right)\right) \tag{9}$$

$$\mathbf{F}_l^{\text{residual1}} = \mathbf{F}_l^{\text{high}} - \mathbf{F}_l^{\text{medium2}} \tag{10}$$

$$\mathbf{F}_l^{\text{down}} = \text{ReLU}\left(\text{Conv}\left(\mathbf{F}_l^{\text{residual1}}\right)\right) + \mathbf{F}_l^{\text{medium1}}. \tag{11}$$

For the up-sampling block [see Fig. 6], $\mathbf{F}_l^{\text{down}}$ is deconvoluted to obtain $\mathbf{F}_l^{\text{medium3}}$, $\mathbf{F}_l^{\text{medium3}}$ is convoluted to obtain $\mathbf{F}_l^{\text{medium4}}$, and the same strategy is used to calculate the residual features ($\mathbf{F}_l^{\text{residual2}}$). The deconvoluted residual features are added with $\mathbf{F}_l^{\text{medium3}}$ to obtain the final upsampling features ($\mathbf{F}_l^{\text{up}}$). The specific formula is given as follows:

$$\mathbf{F}_l^{\text{medium3}} = \text{ReLU}\left(\text{Deconv}\left(\mathbf{F}_l^{\text{down}}\right)\right) \tag{12}$$

$$\mathbf{F}_l^{\text{medium4}} = \text{ReLU}\left(\text{Conv}\left(\mathbf{F}_l^{\text{medium3}}\right)\right) \tag{13}$$

$$\mathbf{F}_l^{\text{residual2}} = \mathbf{F}_l^{\text{down}} - \mathbf{F}_l^{\text{medium4}} \tag{14}$$

$$\mathbf{F}_l^{\text{up}} = \text{ReLU}\left(\text{Deconv}\left(\mathbf{F}_l^{\text{residual2}}\right)\right) + \mathbf{F}_l^{\text{medium3}}. \tag{15}$$

C. Multiscale Low-Rank Deep Back Projection Fusion Network

To combine the multiscale spectral feature extraction model with the multiscale spatial feature extraction network into the Laplacian framework, we propose the MLR-DBPFN for the fusion of noisy HS data and MS data. Fig. 7 shows the flowchart of our method. MLR-DBPFN effectively extracts the spectral features of multisource data (i.e., effectively separates the noise spectral features from the pure spectral features), and the down–up-sampling blocks are used in the process of multiscale spatial feature extraction to ensure the full use of spatial information. Finally, the training process is completed within the Laplacian framework. MLR-DBPFN adaptively reduces the weight of the noise spectral features in the network feedback mechanism, can achieve robust antinoise fusion performance, avoids the parameter sensitivity of traditional methods, and is suitable for the high-quality fusion of different noise data.

First, we superimpose the interpolated LR HS and HR MS to obtain the HSMS and use MLR and convolution operation to extract the multiscale spectral features of the HSMS data. Second, we construct the DBPFN network, use the first layer of spectral features as input data, combine the downsampling block and the upsampling block to extract multiscale spatial features for each layer of spectral features, and add the remaining spectral features to the network in turn. Finally, HR HS is used as the output data for network training.

In our method, the multiscale spectral features can be obtained by formulas (1)–(4). For the DBPFN network, the specific formula is given as follows:

$$\mathbf{F}_{B_1}^{\text{SS}} = \text{Up} \uparrow(\text{Down} \downarrow (\text{ReLU}(\text{Conv}(\mathbf{F}_1)))) \tag{16}$$

$$\mathbf{F}_{B_1} = \text{cat}\left(\mathbf{F}_{B_1}^{\text{SS}}, \text{ReLU}(\text{Conv}(\mathbf{F}_1)), \mathbf{F}_2, 3\right) \tag{17}$$

$$\mathbf{F}_{B_2}^{\text{SS}} = \text{Up} \uparrow \left(\text{Down} \downarrow\left(\text{ReLU}\left(\text{Conv}\left(\mathbf{F}_{B_1}\right)\right)\right)\right)$$

$$\cdots$$

$$\mathbf{F}_{B_n} = \text{cat}\left(\mathbf{F}_{B_{n-1}}^{\text{SS}}, \text{ReLU}\left(\text{Conv}\left(\mathbf{F}_{B_{n-2}}\right)\right), \mathbf{F}_n, 3\right) \tag{18}$$

$$\mathbf{F}_{B_n}^{\text{SS}} = \text{Up} \uparrow \left(\text{Down} \downarrow \left(\text{ReLU}\left(\text{Conv}\left(\mathbf{F}_{B_n}\right)\right)\right)\right) \tag{19}$$

$$\mathbf{F}_{\mathbf{1}} = \text{cat}\left(\mathbf{F}_{B_n}^{\text{SS}}, \text{ReLU}\left(\text{Conv}\left(\mathbf{F}_{B_{n-1}}\right)\right), 3\right) \tag{20}$$

$$\mathbf{F} = \text{ReLU}(\text{Conv}(\mathbf{F}_{\mathbf{1}})) \tag{21}$$

where $\mathbf{F}_{B1}^{\text{SS}}$ represents the feature map of the first layer of spectral features ($\mathbf{F}_1$) after convolution, downsampling block, and upsampling operations, and $\mathbf{F}_{B_1}$ represents the superimposed feature maps. $\mathbf{F}_{B_2}^{\text{SS}}$ represents the feature maps of $\mathbf{F}_{B_1}$ after convolution, downsampling block, and upsampling operations, and $\mathbf{F}_{\mathbf{1}}$ is the superimposed feature maps of $\mathbf{F}_{B_n}^{\text{SS}}$ and convoluted $\mathbf{F}_{B_{n-1}}$. $\mathbf{F}$ is obtained by the convolution of $\mathbf{F}_{\mathbf{1}}$, and $n = 4$ in the experiment. Finally, the l_1-norm is utilized to train MLR-DBPFN with a loss function defined as

$$\text{loss} = \frac{1}{G}\sum_{g=1}^{G}\left\|\mathbf{Z}_r^g - \mathbf{Z}_f^g\right\|_1 \tag{22}$$

where $G \in [1, 2, \ldots, g]$ is the number of training samples, $\mathbf{Z}_r^g$ denotes the gth sample of reference HR HS data, $\mathbf{Z}_f^g$ represents the gth sample of fused data, $\|\cdot\|_1$ is the l_1-norm, and loss is the error between output HR HS data and fused HR HS data.

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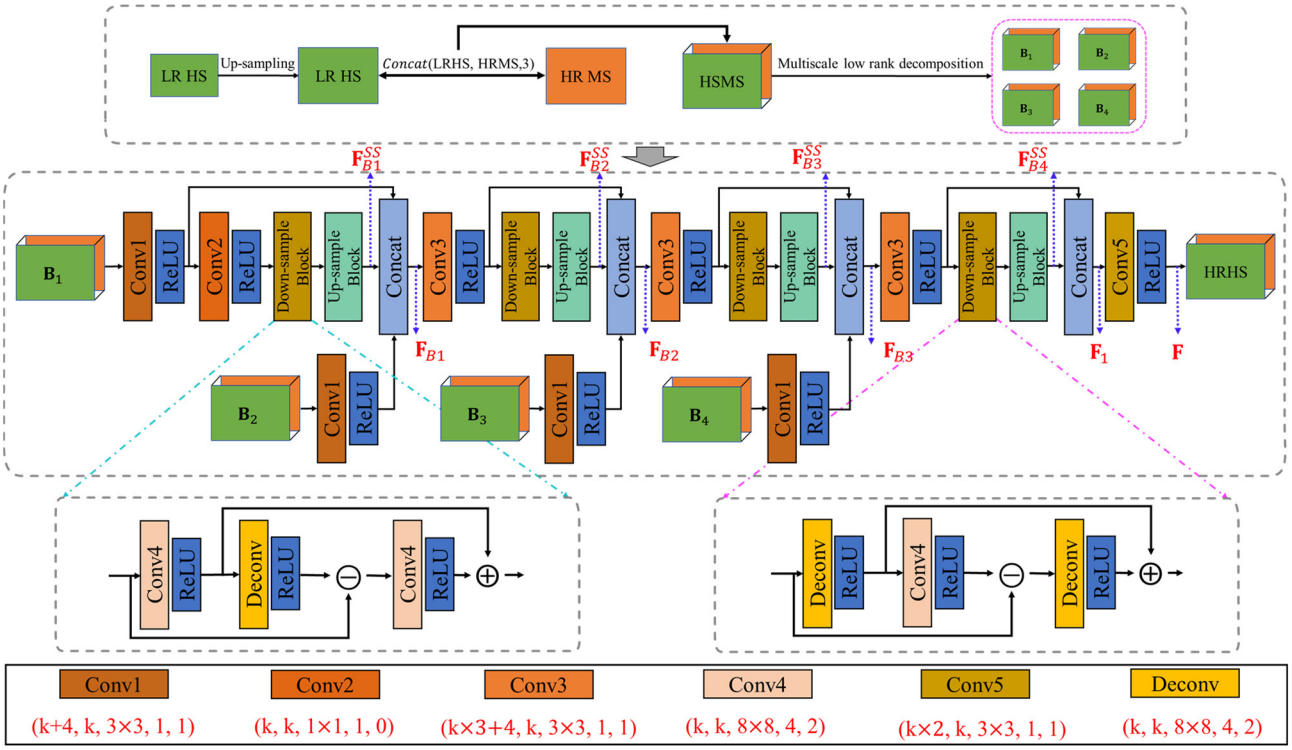


Fig. 7. MLR-DBPFN.

III. Experiments

A. Experimental Data

To verify the fusion performance of our method on noisy HS data, four datasets are used in the experiment. Taking the above datasets as the reference HR HS, we obtain LR HS through spatial downsampling and noise synthesis, and HR MS through spectral downsampling [6]. The specific formula is given as follows:

$$\begin{cases} \mathbf{X_1} = \text{Gaussian}(\mathbf{Z}) \\ \mathbf{X_2} = \text{Bicubic}\left(\mathbf{X_1}, \dfrac{1}{r}\right) \\ \mathbf{X} = \mathbf{X_2} + \mathbf{Noise} \end{cases} \tag{23}$$

$$\mathbf{Y} = \mathbf{R} * \mathbf{Z} \tag{24}$$

where Gaussian$(\cdot)$ is the Gaussian blur operation, Bicubic$(\cdot)$ represents the sampling operation, $\mathbf{X_1}$ is the HS data after performing Gaussian blurring on $\mathbf{Z}$, and $\mathbf{X_2}$ is the HS data after performing spatial downsampling on $\mathbf{X_1}$. r is the ratio of spatial resolution between LR HS and HR MS, **Noise** is the noisy matrix, and $\mathbf{R}$ is the spectral response function. Table I shows the details of the experimental datasets.

PaviaU: The data were captured by the Reflective Optical Spectroscopic Imaging System (ROSIS) sensor, covering a part of the University of Pavia, Italy, which contains 103 bands with a spatial resolution of 1.3 m. The spectral range is 430–860 nm, 98 bands can be used for experiments when removing the bands with serious water vapor absorption, and 600 × 340 pixels were finally used as the reference HS.

Chikusei: The HS data were obtained from the University of Tokyo and collected by the Headwall Hyperspec-VNIR-C sensor in Chikusei, Japan, on July 29, 2014. The data contain 122 bands with 2517 × 2335 pixels, with a spatial resolution of 2.5 m and a spectral range of 343–1018 nm. By eliminating the bands with severe water vapor absorption, 120 bands were used for experiments. Besides, 800 × 800 pixels were used as the reference HS, and 40-dB Gaussian random noise was added to the dataset.

PaviaC: The data are acquired by the ROSIS sensor, covering a part of Pavia, Italy, including 102 bands with 960 × 640 pixels, and the spectral range is 430–860 nm. After excluding the water vapor absorption band, a total of 96 bands were used in the experiment, 800 × 500 pixels were used as the reference HS, and 30% stripe noise was added to the dataset.

DFC2018 Houston: These data are the 2018 IEEE GRSS Data Fusion competition data, produced by Dr. Saurabh Prasad's Laboratory at the University of Houston. The data contains 48 bands with 1-m spatial resolution. In the experiment, 500 × 800 × 48 pixels were used as the reference HS, and mixed noise (40-dB Gaussian random noise + 30% stripe noise) was added to the dataset.

B. Evaluation Metrics

To quantitatively evaluate the fusion performance of our method, the spectral angle measurement (SAM) [1], the Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [6], the peak signal-to-noise ratio (PSNR) [33], and cross correlation (CC) [4] are selected. SAM is used to evaluate the spectral distortion of the fused image, the ideal value is 0, and the larger the value has more serious spectral distortion.

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TABLE I
INFORMATION OF EXPERIMENTAL DATA

Dataset	Reference HS	LR HS	HR MS	Training LR HS	Training HR MS	Testing LR HS	Noise
PaviaU	$600 \times 340 \times 98$	$150 \times 85 \times 98$	$600 \times 340 \times 4$	$32 \times 32 \times 98$	$128 \times 128 \times 4$	$50 \times 50 \times 98$	—
Chikusei	$800 \times 800 \times 120$	$200 \times 200 \times 120$	$800 \times 800 \times 4$	$32 \times 32 \times 120$	$128 \times 128 \times 4$	$50 \times 50 \times 120$	+ 40db gaussian random noise
PaviaC	$800 \times 500 \times 96$	$200 \times 125 \times 96$	$800 \times 500 \times 4$	$32 \times 32 \times 96$	$128 \times 128 \times 4$	$50 \times 50 \times 96$	+ 30% strip noise
DFC2018 Houston	$500 \times 800 \times 48$	$125 \times 200 \times 48$	$500 \times 800 \times 4$	$32 \times 32 \times 48$	$128 \times 128 \times 4$	$50 \times 50 \times 48$	+ 40db gaussian random noise + 30% strip noise

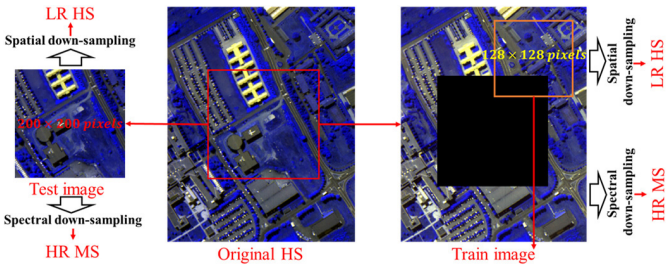


Fig. 8. Example of train images and test images.

ERGAS and PSNR are used for the comprehensive evaluation of spectral and spatial distortions. The larger value the ERGAS and RMSE, the larger the fusion error, and the larger the PSNR value, the better the fusion performance. CC measures spatial distortion of the fused image, and the spatial fidelity is the best when the value is 1.

C. Experimental Settings

The training and testing samples are obtained by cropping the original HS image [see Fig. 8]. We first selected and cropped the test sample area from the original data. Then, the test sample area in the original data is filled with 0, and part of the image is randomly selected for training at each iteration. Therefore, the number of training samples in the experiment is equal to the number of iterations. It is worth noting that there is no overlapping area between the training and testing sample, and we selected $200 \times 200 \times K$ pixels as the test sample and $128 \times 128 \times K$ pixels as the training sample for all datasets. In addition, the Gaussian blur kernel is 3×3 and $r = 4$ during spatial downsampling, and the SPOT-4 spectral response function is used to obtain HR MS for all datasets. To verify the antinoise fusion performance of our method, different types of noise were added to each data. The experiment includes a noise-free dataset, a 40-dB Gaussian random noise dataset, a 30% stripe noise dataset, and a mixed noise (40-dB Gaussian random noise + 30% stripe noise) dataset. It is worth noting that noise exists in each band of the simulated LR HS data, and the distribution of noise in each band is different, which is significantly challenging for image fusion.

MLR-DBPFN decomposes HSMS into four-layer spectral features, uses Adam as the optimizer, and performs 10 000 iterations for all datasets to train the model, and the learning rate is $1e-4$. To avoid dropout and batch norm, DBPFN successively performed a 3×3 convolution operation with a stride $= 1$ and padding $= 1$ on the spectral feature layer, and a 1×1 convolution operation with a stride $= 1$ and padding $= 0$. In the up–down-sampling block, the 8×8 convolution and the deconvolution kernel with a stride $= 4$ and padding $= 2$ are used, which can fully extract the multifrequency spatial information of the data. In addition, the convolution kernel between the up–down-sampling block is 3×3 with stride $= 1$ and padding $= 1$. Nine state-of-the-art methods are selected for comparison, including GSA, MTF-GLP [6], NLSTF [24], CNMF [22], the Fast Fusion based on Sylvester Equation (FUSE) [43], the pyramid fully convolutional network (PFCN) [30], TWO-CNN [29], multiresolution spatial–spectral feature learning (MSSL) [44], and the spatial–spectral reconstruction network (SSRNET) [45]. The parameters of all benchmark methods are derived from the relevant references. Computing equipment contains the 64-GB CPU memory and 1× GPU of GeForce GTX 1080Ti.

D. Experimental Results

Fig. 9 shows the evaluation indicators of MLR-DBPFN of each iteration in four datasets. At 0–1000 iterations, the accuracy improves rapidly. After 1000 iterations, although there are obvious fluctuations in the trend, the overall accuracy improves slowly.

1) Noise-Free Dataset—PaviaU: The PaviaU data are used to evaluate the fusion performance of MLR-DBPFN for the noise-free image. Fig. 10 shows the false-color composite images of the fusion results of different methods. Except for TWO-CNN, all methods have better spectral fidelity. However, GSA, MTF-GLP, FUSE, PFCN, TWO-CNN, and MSSL have spatial distortions at the edges of buildings. NLSTF, CNMF, SSRNET, and our method have consistent visualization effects with the reference image. SAM maps of these fusion results are shown in Fig. 11. GSA, MTF-GLP, FUSE, and TWO-CNN have obvious spectral reconstruction errors, especially in the building area. NLSTF, CNMF, PFCN, MSSL, and SSRNET are followed, and MLR-DBPFN has the smallest error. Table II shows the quantitative evaluation results of the fusion results. GSA and TWO-CNN achieve the worst results among all

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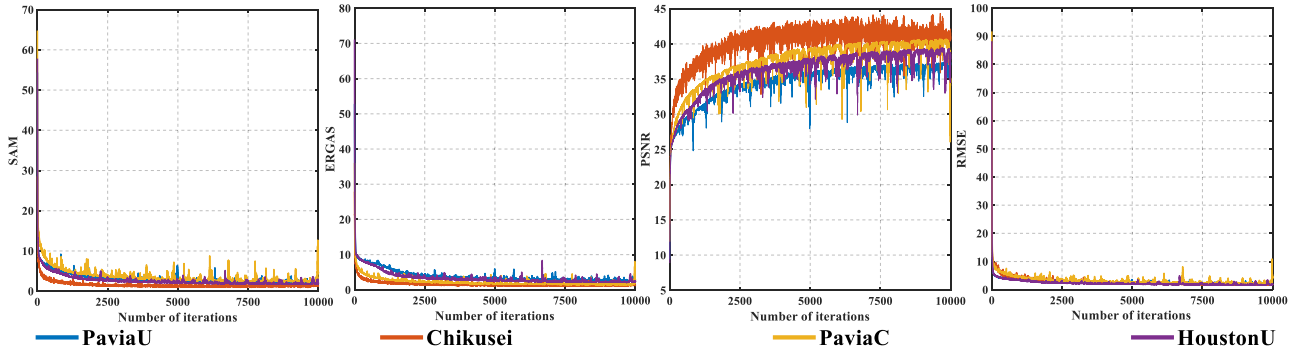


Fig. 9. Model evaluation results obtained by MLR-DBPFN after each iteration.

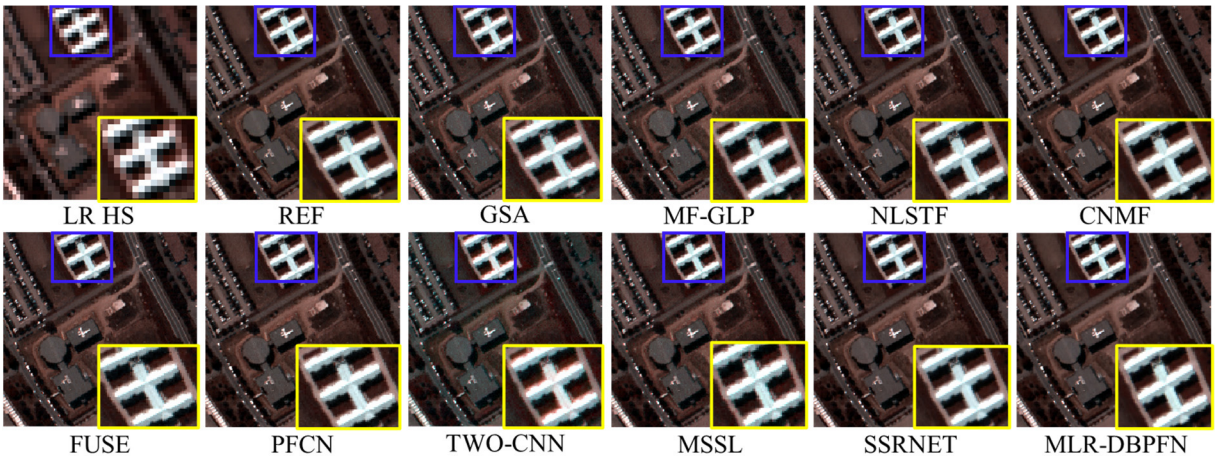


Fig. 10. Fusion results in the noise-free dataset, PaviaU, from different methods.

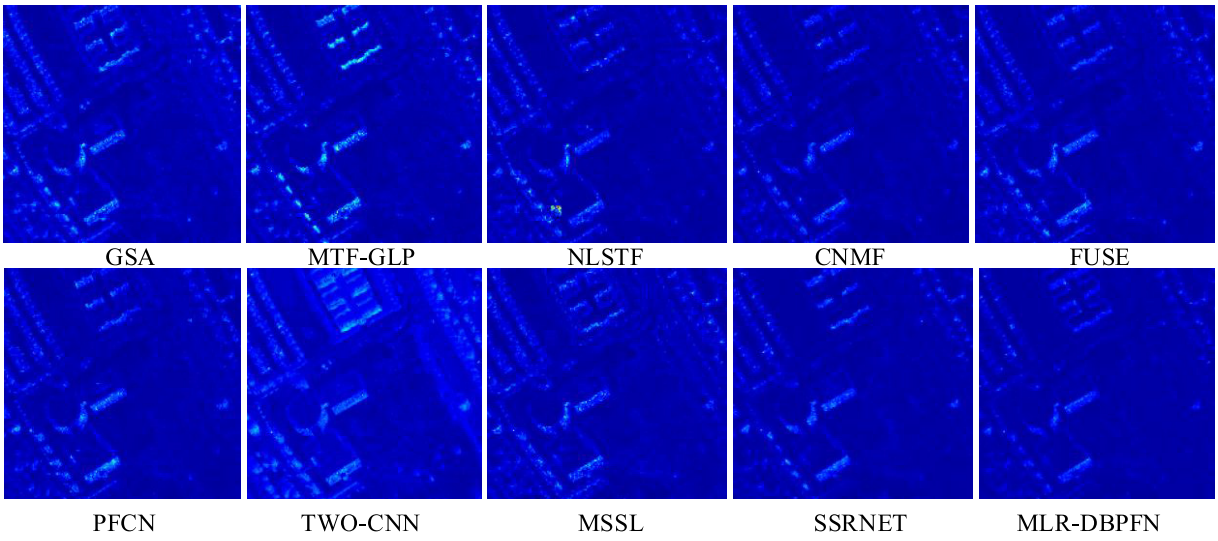


Fig. 11. SAM maps in the noise-free dataset, PaviaU, from different methods.

indicators, other methods have relatively close results, and our method achieves the best value among all indicators.

2) 40-dB Gaussian Random Noise Dataset—Chikusei: To explore the fusion performance of MLR-DBPFN for the HS contaminated by the Gaussian random noise, we added 40-dB Gaussian random noise to the Chikusei dataset, as shown in Fig. 12, LR HS. The fusion results of all methods are shown in Fig. 12. GSA, MTF-GLP, NLSTF, TWO-CNN, and MSSL are more sensitive to the Gaussian random noise, and the fusion results have obvious spectral and spatial distortions. CNMF, FUSE, PFCN, and SSRNET can restrain the noise to a limited extent, but there is still local noise in

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TABLE II
QUANTITATIVE EVALUATION OF THE FUSION RESULTS FOR NOISE-FREE DATA—PAVIAU

Methods	SAM	ERGAS	PSNR	CC
GSA	3.16	2.22	38.42	0.9893
MTF-GLP	2.87	1.94	39.57	0.9921
NLSTF	2.41	1.67	39.53	0.9924
CNMF	2.16	1.37	41.70	0.9954
FUSE	2.45	1.52	41.33	0.9943
PFCN	2.63	1.73	39.31	0.9927
Two-CNN	4.29	3.54	32.88	0.9794
MSSL	2.73	2.01	38.73	0.9908
SSRNET	2.23	1.44	40.89	0.9952
MLR-DBPFN	**2.06**	**1.31**	**42.08**	**0.9961**

TABLE III
QUANTITATIVE EVALUATION OF THE FUSION RESULTS FOR THE 40-dB GAUSSIAN RANDOM NOISE DATASET—CHIKUSEI

Methods	SAM	ERGAS	PSNR	CC
GSA	10.65	11.45	24.87	0.8749
MTF-GLP	11.09	9.82	24.66	0.8759
NLSTF	14.10	16.33	21.98	0.7954
CNMF	7.36	5.43	28.26	0.9372
FUSE	5.77	6.18	30.27	0.9564
PFCN	4.66	3.94	32.32	0.9714
Two-CNN	12.03	13.24	23.47	0.8121
MSSL	12.99	14.44	22.94	0.8348
SSRNET	4.58	4.50	31.93	0.9699
MLR-DBPFN	**2.53**	**3.48**	**37.19**	**0.9843**

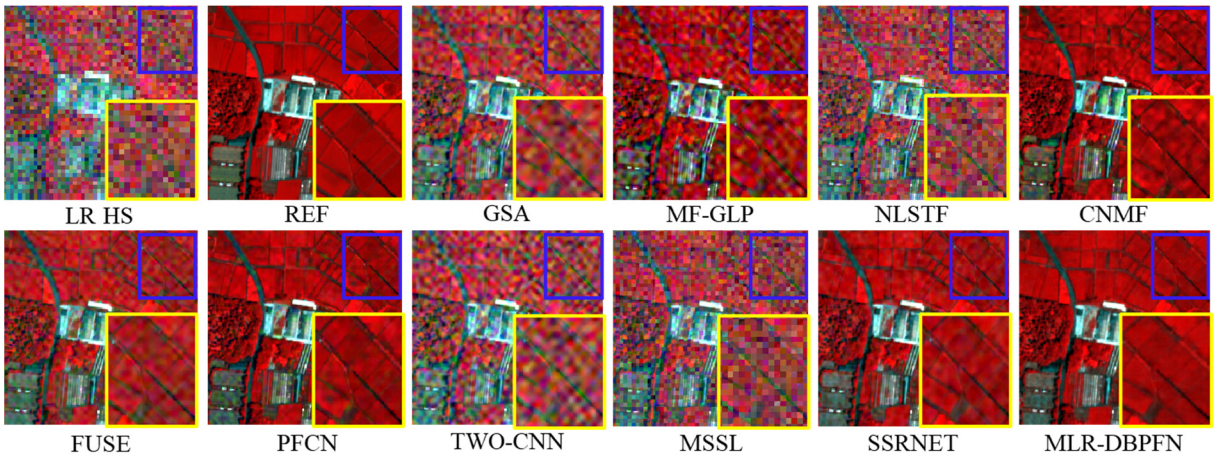


Fig. 12. Fusion results in the 40-dB Gaussian random noise dataset, Chikusei, from different methods.

the fused images. MLR-DBPFN is almost the same as the reference image in terms of visualization. Fig. 13 shows SAM maps of these fusion results. GSA, MTF-GLP, NLSTF, TWO-CNN, and MSSL have global spectral reconstruction errors. Spectral reconstruction errors of CNMF, FUSE, PFCN, and SSRNET are the second, mainly in water bodies. Our method achieves the best spectral fidelity. Table III shows the quantitative evaluation results of various methods. GSA, MTF-GLP, NLSTF, TWO-CNN, and MSSL have poor values, and the fused images have serious errors. CNMF, FUSE, PFCN, and SSRNET followed. Our method has obvious antinoise performance for the image contaminated by the Gaussian random noise and achieves the best results in all indicators.

3) 30% Stripe Noise Dataset—PaviaC: To verify the fusion performance of MLR-DBPFN for the image contaminated by stripe noise, we added 30% stripe noise to the PaviaC data, as shown in Fig. 14, LR HS. The false-color composite images of the fusion results of all methods are shown in Fig. 14. Except for SSRNET and our method, other benchmark methods are significantly affected by stripe noise and have severe spectral distortion, and NLSTF has obvious spatial errors compared with the reference image. Fig. 15 shows the SAM maps of these fusion results. GSA, MTF-GLP, NLSTF, CNMF, FUSE, PFCN, TWO-CNN, and MSSL have obvious stripe-like spectral reconstruction errors, which shows that these methods are seriously affected by stripe noise and cannot suppress

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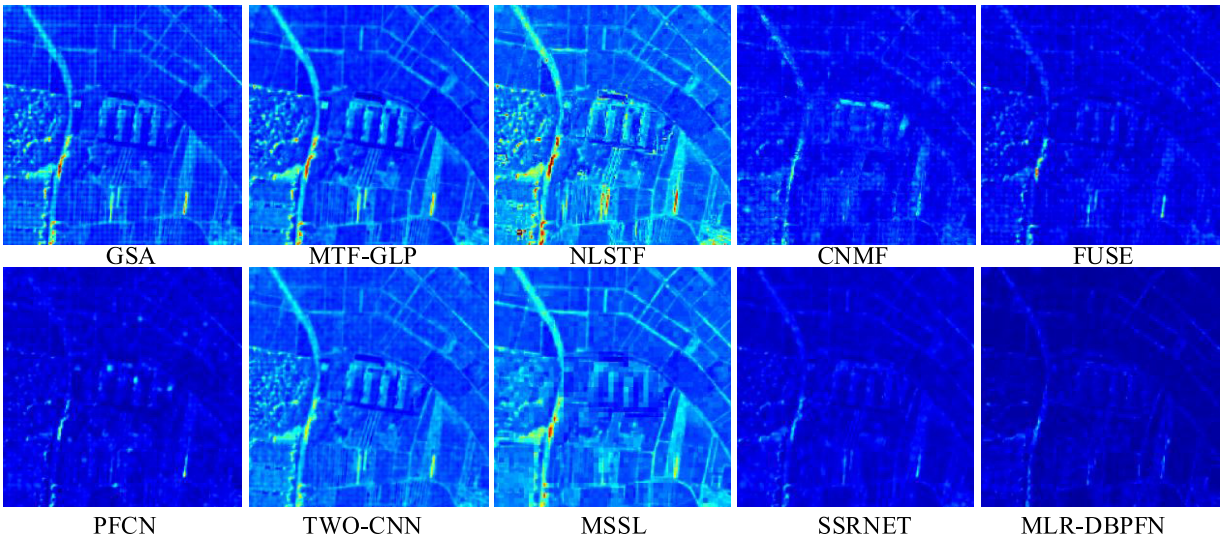


Fig. 13. SAM maps in the 40-dB Gaussian random noise dataset, Chikusei, from different methods.

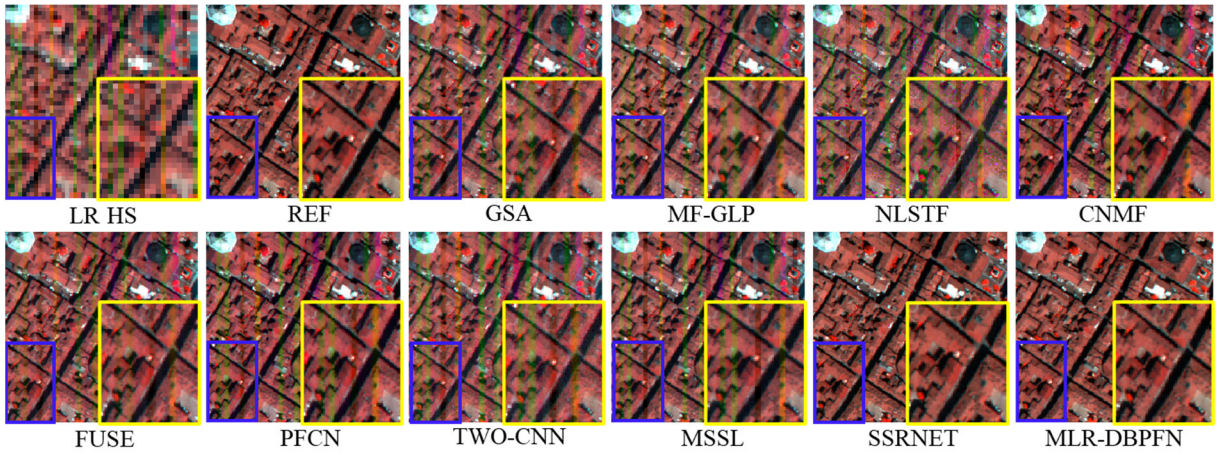


Fig. 14. Fusion results in the 30% stripe noise dataset, PaviaC, from different methods.

stripe noise in LR HS. Although there is no stripe-like error in SSRNET, the spectral reconstruction error is still obvious compared with our method. Table IV shows the quantitative evaluation values of the fusion results. MLR-DBPFN has the best results among all indicators, followed by SSRNET, and other methods have serious spectral and spatial distortions, which shows that MLR-DBPFN can effectively suppress stripe noise while still achieving better fusion performance.

4) Mixed Noise Dataset—DFC2018 Houston: Mixed noise is more common in LR HS images. We added 40-dB Gaussian random noise and 30% stripe noise to the DFC2018 Houston dataset, as shown in Fig. 16-LR HS. GSA, MTF-GLP, FUSE, PFCN, TWO-CNN, and MSSL have obvious spatial distortions. CNMF and SSRNET have better spatial resolution enhancement effects. However, the spectral distortion is serious. Our method achieves the most consistent visualization results with reference effects. As shown in Fig. 17, our method achieves the best spectral fidelity, followed by SSRNET. However, other methods are very sensitive to mixed noise and have serious spectral reconstruction errors. Table V shows quantitative evaluation results of various methods. All benchmark methods have serious spectral and spatial distortions and have extremely poor quantitative evaluation results. Our method consistently maintains the best results, which indicates that MLR-DBPFN can effectively suppress mixed noise and has robust fusion performance for LR HS contaminated by different noises.

E. Discussion

To determine the optimal number of decomposition layers of spectral feature (L), we used different L to enter MLR-DBPFN. In the experiment, the range of L is 2–10, the stride is 1, and the number of iterations is set to 10 000. The quantitative evaluation results are shown in Table VI, and Fig. 18 shows the PSNR and computational time of different L's in MLR-DBPFN. For all datasets, the fusion performance increases rapidly when $2 \leq L \leq 4$, and there are slight fluctuations and a generally stable trend when

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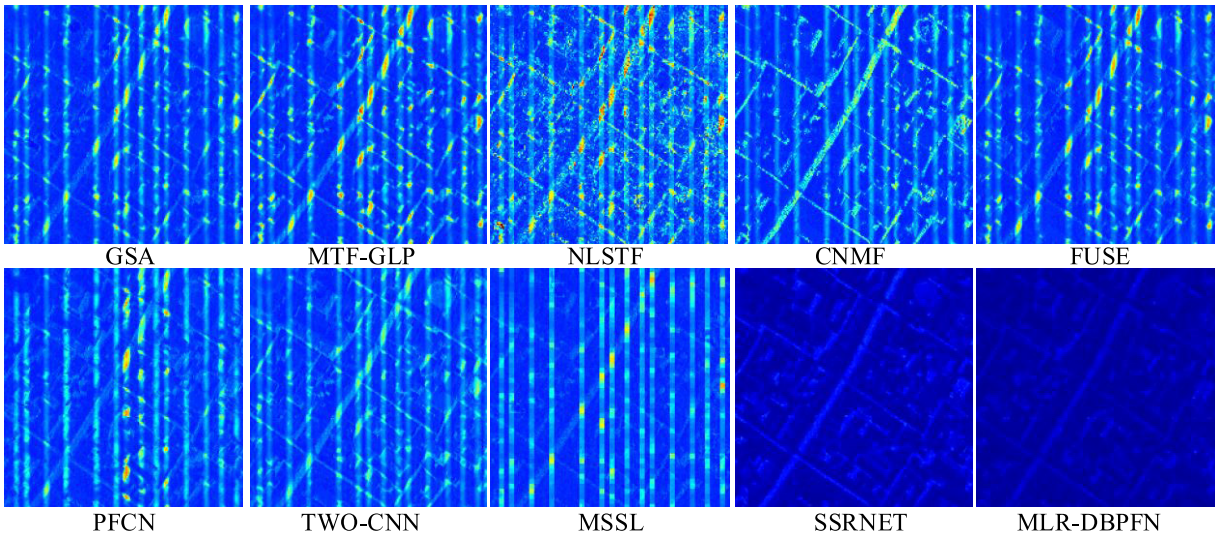


Fig. 15. SAM maps in the 30% stripe noise dataset, PaviaC, from different methods.

TABLE IV
QUANTITATIVE EVALUATION OF THE FUSION RESULTS FOR THE 30% STRIPE NOISE DATASET—PAVIAC

Methods	SAM	ERGAS	PSNR	CC
GSA	11.76	12.42	12.94	0.9500
MTF-GLP	12.39	13.36	12.91	0.9578
NLSTF	14.25	15.15	12.93	0.9024
CNMF	12.22	13.34	12.94	0.9552
FUSE	11.84	12.31	12.89	0.9641
PFCN	11.98	12.20	13.05	0.9457
Two-CNN	11.94	12.64	12.93	0.9223
MSSL	11.59	18.66	13.54	0.9486
SSRNET	2.99	1.57	40.16	0.9955
MLR-DBPFN	**1.72**	**1.28**	**42.65**	**0.9963**

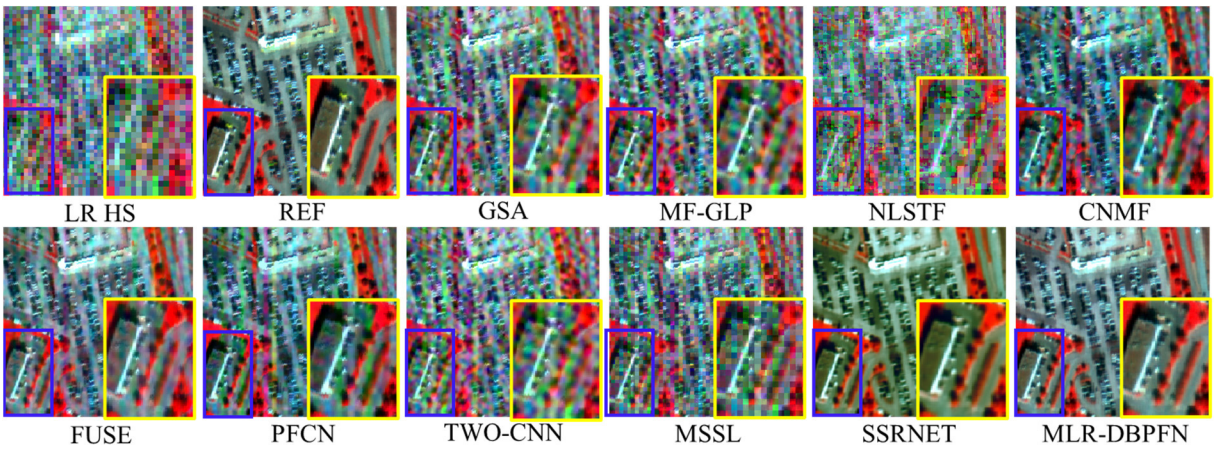


Fig. 16. Fusion results in the mixed noise dataset, DFC2018 Houston, from different methods.

$L > 4$. The reason is that more L means that more iterations are needed to achieve better fusion quality. Therefore, when L keeps increasing, the number of iterations does not increase, which may reduce the fusion performance. However, with the increase in L, hyperparameters continue to accumulate, and the computational efficiency decreases rapidly. To balance fusion performance and computational efficiency, L is selected as 4.

In addition, we also verified that MLR is crucial for MLR-DBPFN. The experiment superimposes LR HS and HR MS data as input data (without extracting multiscale spectral features), and only DBPFN is used for fusion. Fig. 19 shows

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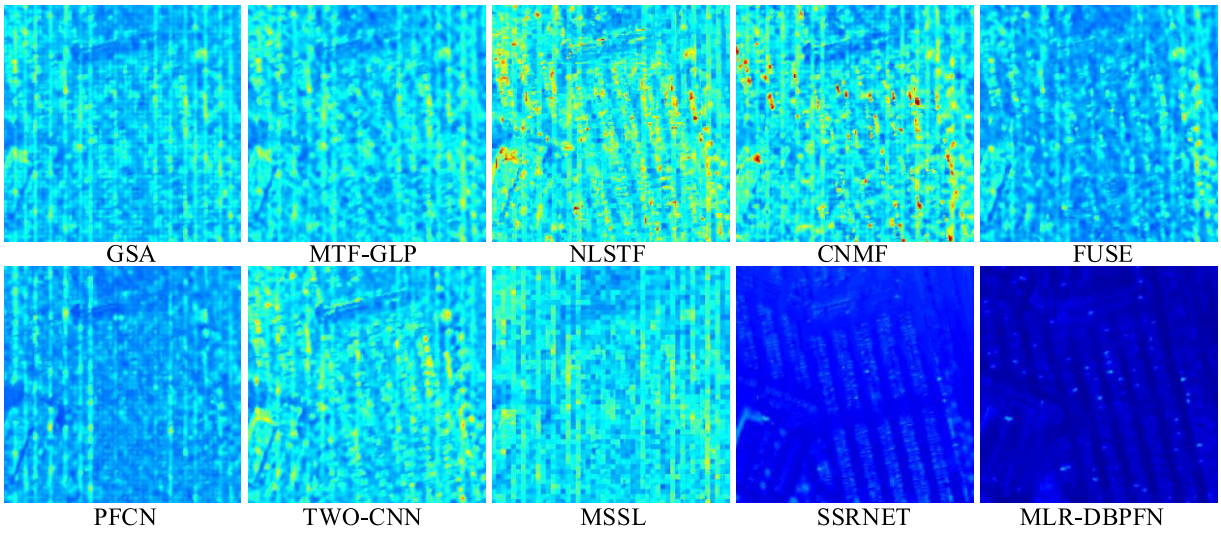


Fig. 17. SAM maps in the mixed noise dataset, DFC2018 Houston, from different methods.

TABLE V

QUANTITATIVE EVALUATION OF THE FUSION RESULTS FOR THE MIXED NOISE DATASET—DFC2018 HOUSTON

Methods	SAM	ERGAS	PSNR	CC
GSA	18.80	18.02	9.47	0.8231
MTF-GLP	19.53	20.31	9.46	0.8033
NLSTF	22.65	25.60	9.15	0.7133
CNMF	21.43	24.32	9.45	0.8014
FUSE	17.57	17.51	10.21	0.8799
PFCN	16.54	17.96	11.06	0.8721
Two-CNN	20.23	22.74	10.06	0.7559
MSSL	20.27	22.73	9.99	0.7738
SSRNET	7.60	4.49	27.92	0.9199
MLR-DBPFN	**2.94**	**1.93**	**32.80**	**0.9628**

TABLE VI

QUANTITATIVE EVALUATION OF DIFFERENT DECOMPOSITION NUMBERS OF SPECTRAL FEATURES OF MLR-DBPFN

Data	Method	Decomposition Number of Spectral Feature								
		2	3	4	5	6	7	8	9	10
PaviaU	SAM	2.57	2.42	2.06	2.06	2.08	2.06	**2.05**	2.06	2.09
	ERGAS	1.50	1.44	1.31	1.29	1.35	**1.27**	1.29	1.38	1.56
	PSNR	40.81	40.92	**42.08**	41.59	40.01	42.55	41.69	39.97	39.58
	CC	0.9945	0.995	**0.996**	0.995	**0.996**	0.995	0.991	0.995	0.991
	Training time (min)	7	9	15	30	58	87	116	143	181
Chikusei	SAM	7.65	5.32	2.53	**2.52**	2.53	2.56	**2.52**	2.53	2.57
	ERGAS	9.02	7.24	**3.48**	3.54	3.51	3.59	4.26	3.91	3.56
	PSNR	28.60	30.64	**37.19**	35.24	36.94	37.16	37.13	36.89	38.56
	CC	0.916	0.933	**0.984**	**0.984**	0.962	0.966	**0.984**	**0.984**	0.962
	Training time (min)	10	14	19	36	57	83	106	134	170
PaviaC	SAM	8.21	5.32	1.72	1.71	1.76	**1.69**	1.70	1.73	1.74
	ERGAS	500.24	60.89	**1.28**	1.56	2.89	1.64	3.25	1.89	1.31
	PSNR	28.34	34.26	42.65	39.21	37.51	41.72	**43.08**	40.69	42.01
	CC	0.983	0.989	**0.996**	**0.996**	0.994	**0.996**	**0.996**	0.991	0.989
	Training time (min)	18	22	25	37	62	83	114	138	166
HoustonU	SAM	15.26	8.95	**2.94**	**2.94**	2.95	3.93	3.94	4.96	3.97
	ERGAS	435.16	15.69	**1.93**	5.84	4.16	1.99	1.84	3.46	2.85
	PSNR	13.62	28.16	**32.80**	29.52	28.54	31.59	33.15	31.85	30.26
	CC	0.925	0.930	0.962	**0.963**	0.962	0.961	**0.963**	**0.963**	0.962
	Training time (min)	15	20	27	43	67	72	103	140	161

the experimental results and SAM maps. For the noise-free dataset, DBPFN obtains similar fusion performance to MLR-DBPFN, and the spectral distortion mainly exists in the building area. However, for the Gaussian random noise, stripe noise, and mixed noise datasets, DBPFN has obtained obvious reconstruction errors compared with MLR-DBPFN, and the

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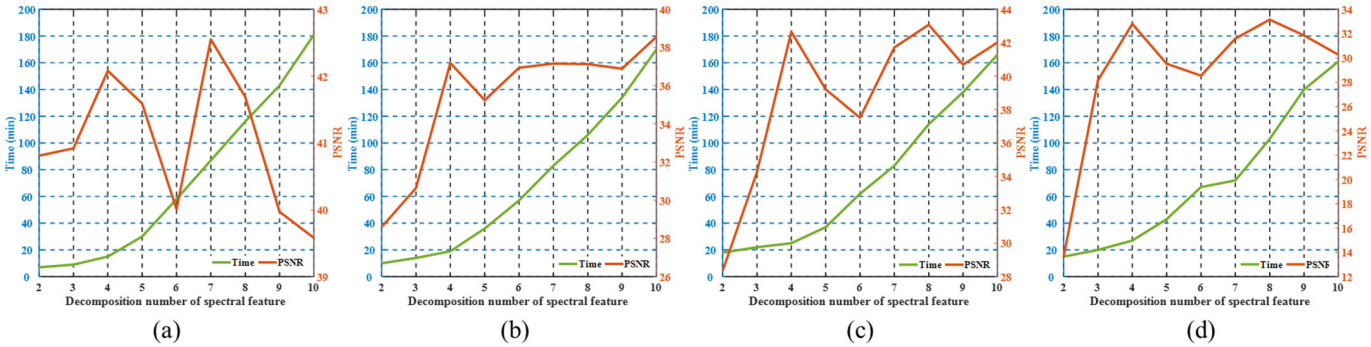


Fig. 18. Comparative results of decomposition number of spectral features on PSNR and time. (a) PaviaU dataset. (b) Chikusei dataset. (c) PaviaC dataset. (d) DCF2018 HoustonU.

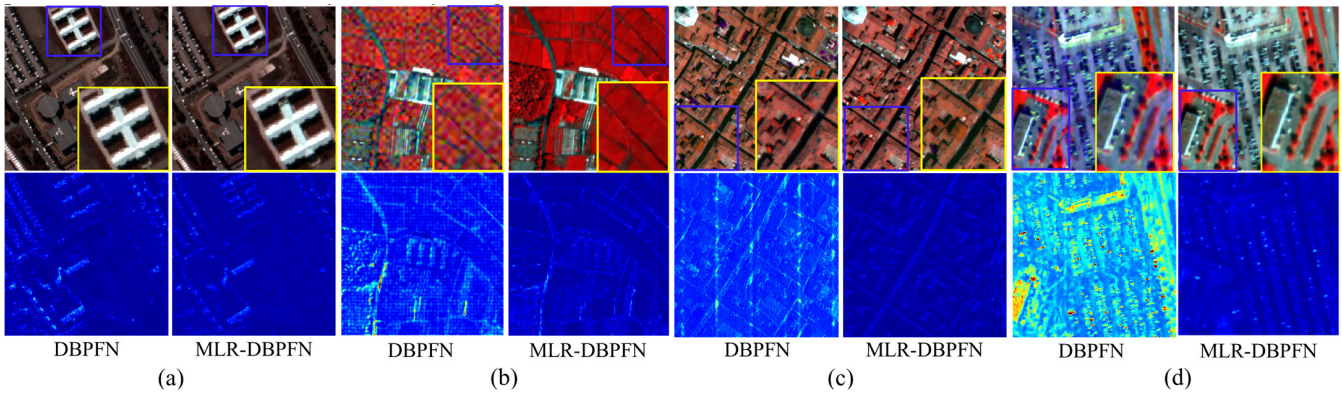


Fig. 19. Comparative results of DBPFN and MLR-DBPFN. The first row represents the fusion result, and the second row represents the SAM graph. (a) PaviaU dataset. (b) Chikusei dataset. (c) PaviaC dataset. (d) DCF2018 HoustonU.

TABLE VII
QUANTITATIVE EVALUATION OF THE FUSION RESULTS OF DBPFN AND MLR-DBPFN

Methods	PaviaU		Chikusei		PaviaC		DCF2018 HoustonU	
	DBPFN	MLR-DBPFN	DBPFN	MLR-DBPFN	DBPFN	MLR-DBPFN	DBPFN	MLR-DBPFN
SAM	2.57	**2.06**	8.78	**2.53**	10.15	**1.72**	19.68	**2.94**
ERGAS	1.55	**1.31**	9.49	**3.48**	1815.06	**1.28**	3441.41	**1.93**
PSNR	40.69	**42.08**	26.70	**37.19**	12.91	**42.65**	9.47	**32.80**
CC	0.9942	**0.9961**	0.9157	**0.9843**	0.9832	**0.9963**	0.9003	**0.9628**

spectral distortion mainly exists in the noise distribution area. Table VII shows quantitative evaluation results of DBPFN and MLR-DBPFN in different datasets. DBPFN cannot suppress noise during the fusion process, and MLR is necessary for noisy image fusion.

IV. CONCLUSION

In this article, an effective and robust antinoise HS fusion network MLR-DBPFN is presented. Our method can decompose complex spectral features into multiple simple spectral components while separating noisy spectral features, avoiding local reconstruction errors caused by global spectral features in the fusion process. Besides, the spatial fidelity of the reconstructed data is guaranteed by combining it with a multiscale spatial feature extraction network. The network feedback mechanism is used to reduce the weight of noisy spectral features to effectively suppress noise and realize the high-quality fusion of noisy HS. Experimental results prove that MLR-DBPFN can effectively remove noise while obtaining comparative fusion performance. However, MLR-DBPFN only conducts experiments on datasets with a spatial resolution ratio of 4. How to make improvements to make it suitable for fusion with a large spatial resolution ratio is one of our future research directions.

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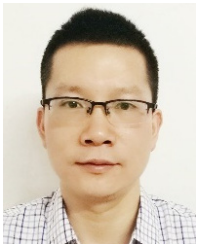
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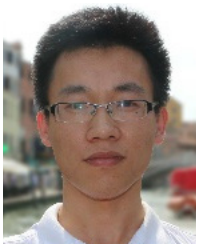
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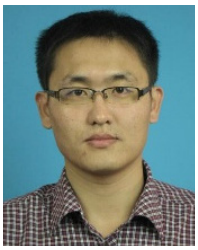
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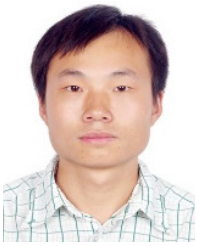
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